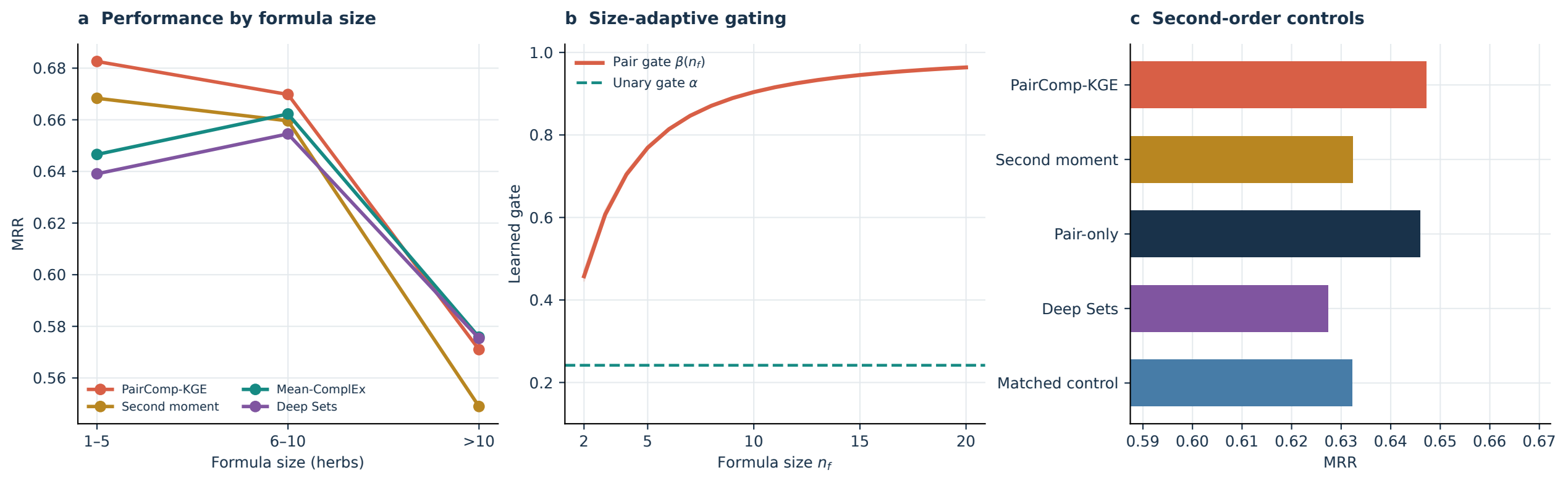
a Performance by formula size
MRR
0.68
0.66
0.64
0.62
0.60
0.58
0.56
PairComp-KGE
Second moment
Mean-ComplEx
Deep Sets
1–5
6–10
>10
Formula size (herbs)
b Size-adaptive gating
Learned gate
1.0
0.8
0.6
0.4
0.2
Pair gate $\beta(n_f)$
Unary gate α
2
5
10
15
20
Formula size n_f
c Second-order controls
PairComp-KGE
Second moment
Pair-only
Deep Sets
Matched control
0.59
0.60
0.61
0.62
0.63
0.64
0.65
0.66
0.67
MRR